Vector Spaces: A Quick Recap

- Before going further with orthogonality, it is worth recalling the general algebraic framework it lives in: vector spaces and inner products.
- A **vector space** V over $\mathbb{R}$ is a set equipped with two operations:
  - **addition**: $v + w \in V$ for $v, w \in V$;
  - **scalar multiplication**: $cv \in V$ for $c \in \mathbb{R},\ v \in V$;

  satisfying the usual axioms (associativity, distributivity, a zero vector, etc.).
- A vector space has a **dimension**, which need not be finite.
- The set of all polynomials in x, denoted $\mathcal{P}(x)$, is an infinite-dimensional vector space — exactly the space we have been working in throughout this chapter.



Inner Product Spaces

- A **bilinear form** on V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ that is linear in both slots:

$$\langle au+v, w \rangle = a\langle u, w \rangle + \langle v, w \rangle, \qquad \langle u, av+w \rangle = a\langle u, v \rangle + \langle u, w \rangle.$$

- An **inner product** on V is a bilinear form that is also
  - **symmetric**: $\langle v, w \rangle = \langle w, v \rangle$;
  - **positive definite**: $\langle v, v \rangle \geq 0$, with equality iff $v = 0$.
- A vector space equipped with an inner product is an **inner product space**.
- The Legendre inner product $\langle f, g \rangle = \int_{-1}^{1} f(x)g(x)\,dx$ used throughout this section is exactly an instance of this general definition, on $V = \mathcal{P}(x)$.



The Norm of a Vector

- For v in an inner product space, define its **norm** (length) by

$$\|v\| := \langle v, v \rangle^{1/2}.$$

- The norm satisfies, for all $v, w \in V$ and $a \in \mathbb{R}$:
  - $\|0\| = 0$, and $\|v\| > 0$ if $v \neq 0$;
  - $\|v + w\| \leq \|v\| + \|w\|$ (triangle inequality);
  - $\|av\| = |a| \, \|v\|$.
- This is exactly the norm $\|P_n(x)\|_2 = \sqrt{\langle P_n, P_n \rangle}$ already computed for Legendre polynomials — now recognized as a special case of this general definition.



Orthogonal and Orthonormal Functions

- A sequence of functions $\{y_n(x), n = 1, 2, 3, \dots\}$ on $[a, b]$ is **orthogonal** if

$$\langle y_n, y_m \rangle = \int_a^b y_n(x) y_m(x)\, dx = \begin{cases} 0 & n \neq m, \\ \alpha_n \neq 0 & n = m. \end{cases}$$

- If $\alpha_n = 1$ for all n, the functions are **normalized**, and the sequence is **orthonormal**.
- More generally, $\{y_n(x)\}$ is orthogonal with respect to a weight function $w(x) > 0$ if

$$\langle y_n, y_m \rangle = \int_a^b w(x) y_n(x) y_m(x)\, dx = \begin{cases} 0 & n \neq m, \\ \alpha_n \neq 0 & n = m. \end{cases}$$

- The norm of y_n is $\|y_n\|_2 = \sqrt{\langle y_n, y_n \rangle}$.



Examples of Orthogonal Systems

- $\{y_n(x) = \sin nx,\ \ n = 1, 2, 3, \dots\}$ is orthogonal on $[-\pi, \pi]$.
- $\{y_n(x) = \cos nx,\ \ n = 0, 1, 2, \dots\}$ is orthogonal on $[-\pi, \pi]$.
- $\{1, \sin nx, \cos nx,\ \ n = 1, 2, \dots\}$ is orthogonal on $[-\pi, \pi]$.
- $\left\{\dfrac{1}{\sqrt{2\pi}}, \dfrac{\cos nx}{\sqrt{\pi}}, \dfrac{\sin nx}{\sqrt{\pi}},\ \ n = 1, 2, \dots\right\}$ is **orthonormal** on $[-\pi, \pi]$ (each element divided by its norm).
- If a linearly independent set is not orthogonal, the **Gram–Schmidt process** produces an orthogonal set from it.
- If $\{y_n(x)\}$ is orthogonal (or orthonormal) on $[a, b]$, any f on $[a, b]$ can be expanded in a series of the y_n — giving a Fourier-type series for that family.



Orthogonality of Legendre Polynomials: Setup

- **Prove the orthogonal results of Legendre polynomials.**
- $P_m(x)$ satisfies Legendre's equation with $p = m$, and $P_n(x)$ with $p = n$; writing both in self-adjoint (Sturm–Liouville) form:

$$\left[(1-x^2)P_m'(x)\right]' + m(m+1)P_m(x) = 0,$$

$$\left[(1-x^2)P_n'(x)\right]' + n(n+1)P_n(x) = 0.$$

- Multiply the first by P_n, the second by P_m, subtract, and integrate over $[-1, 1]$:

$$\int_{-1}^{1} \Big(P_n[(1-x^2)P_m']' - P_m[(1-x^2)P_n']' \Big)\, dx$$

$$+ \left[m(m+1) - n(n+1)\right] \int_{-1}^{1} P_m P_n \, dx = 0.$$



Orthogonality Proof: $m \neq n$

- Evaluate the first integral by parts:

$$\int_{-1}^{1} \Big(P_n[(1-x^2)P_m']' - P_m[(1-x^2)P_n']' \Big)\, dx$$

$$= \Big[P_n(1-x^2)P_m' \Big]_{-1}^{1} - \Big[P_m(1-x^2)P_n' \Big]_{-1}^{1}$$

$$- \int_{-1}^{1} \Big(P_n'(1-x^2)P_m' - P_m'(1-x^2)P_n' \Big)\, dx.$$

- The boundary terms vanish since $(1-x^2) = 0$ at $x = \pm 1$; the remaining integral is identically 0 (the integrand is 0).
- So the whole first integral is 0, leaving

$$\big[m(m+1) - n(n+1) \big] \int_{-1}^{1} P_m P_n \, dx = 0.$$

- If $m \neq n$, the bracket is nonzero, so $\displaystyle\int_{-1}^{1} P_m(x) P_n(x)\, dx = 0.$



The Norm of $P_n(x)$, Method 1: Via Rodrigues' Formula

- What if $m = n$? We use Rodrigues' formula and repeated integration by parts:

$$\int_{-1}^{1} [P_n(x)]^2\, dx = \int_{-1}^{1} P_n(x)\, \frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2-1)^n\, dx$$

$$= \frac{1}{2^n n!} \left[P_n(x) \tfrac{d^{n-1}}{dx^{n-1}}(x^2-1)^n \right]_{-1}^{1}$$

$$- \frac{1}{2^n n!} \int_{-1}^{1} P_n'(x) \tfrac{d^{n-1}}{dx^{n-1}}(x^2-1)^n\, dx.$$

- The boundary term vanishes since $(x^2-1)^n$ and its first $n-1$ derivatives all vanish at $x = \pm 1$.
- Integrating by parts n-times(each time the boundary term vanishes):

$$\int_{-1}^{1} [P_n(x)]^2\, dx = \frac{(-1)^2 n}{2^n n!} \int_{-1}^{1} [P_n(x)]^{(n)} (x^2-1)^n\, dx$$

$$= \frac{(-1)^n}{2^n n!} \int_{-1}^{1} \frac{(2n)!}{2^n n!} (1-x^2)^n\, dx.$$



Method 1: Completing the Calculation

- So $\int_{-1}^{1} [P_n(x)]^2\,dx = \dfrac{2(2n)!}{2^{2n}(n!)^2} \int_0^1 (1-x^2)^n\,dx$ (using that $(1-x^2)^n$ is even).
- Substitute $x = \sin\theta$,

$$\int_0^1 (1-x^2)^n\,dx = \int_0^{\pi/2} \cos^{2n+1}\theta\,d\theta = \frac{2n}{2n+1}\cdot\frac{2n-2}{2n-1}\cdots\frac{2}{3}\cdot 1.$$

- Putting this together (Wallis-type reduction):

$$\int_{-1}^{1} [P_n(x)]^2\,dx = \frac{2(2n)!}{2^{2n}(n!)^2}\cdot\frac{2n}{2n+1}\cdot\frac{2n-2}{2n-1}\cdots\frac{2}{3} = \frac{2}{2n+1}.$$

- (OR) Use $\int_0^{\pi/2} \sin^p x\,\cos^q x\,dx = \dfrac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)} \qquad (p, q > -1).$
- Hence

$$\boxed{\int_{-1}^{1} [P_n(x)]^2\,dx = \frac{2}{2n+1}}$$

